5. Project Executing, Monitoring & Controlling

Q.1 Why meetings are useful in Project Monitoring? What rules should be followed to maximize the effectiveness of meetings?

⇒

- Project monitoring is the integral part of the project management.
- It provides the understanding of the progress of the projects so that appropriate corrective actions can be taken.
- Meetings are essential in project monitoring for several reasons:

1) Communication
⇒ Meetings facilitate the exchange of information, updates and feedback among team members and stakeholders.

2) Decision Making
⇒ They provide a platform for making critical decisions, resolving issues and setting priorities.

3) Accountability
⇒ Meetings help in holding team members accountable for their tasks and progress.

4) Alignment
⇒ They ensure that everyone is aligned with the project's goals, objectives and timelines.

- Rules to maximize Meeting Effectiveness:

a) Set clear objectives
⇒ Define the purpose of the meetings and what needs to be achieved.

b) Prepare an agenda
⇒ Share the agenda in advance, outlinging topics to be discussed and time allocated to each.

c) Invite relevant participants
⇒ Only include individuals whose presence is necessary for the topics being discussed.

d) Time management
⇒ Start and end the meeting on time to respect participant's schedule.

e) Encourage Participation
⇒ Create an environment where all participants feel comfortable sharing their thoughts and ideas.

F) Assign action items
⇒ Clearly define action items, responsibilities and deadlines for followup.

g) Follow-up

⇒ Send a summary of the meeting outcomes and action items to all participants.

- Adhering to these rules can help maximize the effectiveness of project monitoring meetings.

Q.2 How communication is planned and managed in Project management?

⇒

- Excellent communication is a critical component of a project success.
- In fact, according to the PMI, most project failures ~~occur~~ are due to communication problems.

a) Plan Project Communication Management:

- The first step is to plan how will you manage communications on your project and across all of your stakeholders.
- This is done by creating a project communications management plan.
- The following factors are addressed.

i) Audience

⇒ This is a list of all stakeholders affected by the project.

- It should include team members, sponsors, customers and other interested parties.

ii) Objective

⇒ Identify the objective of your communications.

- You ~~can~~ may use some communications for awareness, like a status report, and others that require action.

iii) Message

⇒ Decide and create message for each type of communication.

- This is the actual content that will be shared.
- Key components to be communicated include scope, schedule, budget, objectives, risks and deliverables.

iv) Channel

⇒ Choose the right channel through which the message will be delivered.

- "Will it be a formal report emailed out?"

- Your communication plan should be detailed enough to lay out exactly what will be sent, to whom, how, when and who is responsible.

b) Manage Project Communications

- Once the plan has been created and approved, it's the project manager's job to ensure it's managed well.
- This means that the plan needs to be reviewed and updated on a regular basis to reflect any changes to the project or its stakeholders.
- The project manager also has to manage the execution of the communication management plan.

- This includes:

i) Collection and analysis of data
ii) Creation of messages for communication
iii) Transmission or distribution of communication
iv) Storage of any communication reports, files or documents
v) Retrieval of any stored communications
vi) Disposal of any old communications upon project closure or a set date.

c) Monitor Project Communication

- This process is called control communication
- It involves monitoring and controlling project communications throughout the lifecycle
- This may include:

i) Confirming they are received by the proper stakeholder
ii) Confirming messages are understood.
iii) Confirming any relevant feedback is provided to the appropriate project members.

Q.3 What are the importance of vendor documents? How the vendor documents should be preserved?

⇒

- Vendor documents are all the official records and papers shared between a company and its vendors.
- These may include contracts, invoices, quotations, delivery notes, quality certificates and correspondance.
- Importance:

1) Legal Protection
⇒ These documents serve as proof of agreements and transactions.
- In case of disputes, they protect both the company and the vendor.

2) Audit and compilances
⇒ Required during internal and external audits.
- Ensures that company follows proper financial and legal practices.

3) Tracking deliveries and Payments
⇒ Help monitor what was ordered, what was received, and whether it was paid.

- Ensures accuracy in payments and reduces fraud.

4) Performance Evaluation
⇒ Help track vendor performance over time.
- Useful for making future business decisions.

5) Warranty and Support
⇒ Some vendor documents contain warranty info or service agreements.
- Importance of claiming repairs, replacement or support.

• How should be vendor documents Preserved?

- Proper ~~pos~~ preservation of vendor documents ensures that they are safe, accessible and usable when needed.

1) Organized storage
⇒ Store in labeled folders or binders.
- Save digital copies in organized folders on secured servers or cloud systems.

2) Backup and Redundancy
⇒ Always keep backup copies in case originals are lost or damaged.

3) Security and Access control
⇒ Restrict access to only authorized personnel.
- Use passwords and encryption for digital records.

4) Proper Indexing
⇒ Use tags or references numbers for easy search and retrieval.

5) Regular reviews and updates
⇒ Review documents regularly to remove outdated records.
- Update records when contracts are renewed or modified.

6) Compilance with Retention Policies
⇒ Follow legal and organizational rules for how long to keep documents.

7) Disaster Recovery planning
⇒ Have a plan for recovering documents in case of fire, cyber attack or system crash.

Q.4 Explain the Project management template with a sample template sheet.

=>

- A project management template is a pre-designed document that help organize and plan various aspects of a project.
- It serves as a structured framework to outline tasks, timelines, resources and responsibilities.
- Key Components :

1) Project overview
=> Provides a brief summary of a project, including its purpose, objectives and scope.

2) Project scope
=> Defines the boundaries of the project, detailing what is included and what is excluded.

3) WBS
=> Breaks down the project into smaller, manageable tasks or work packages.

4) Scheduled Management Plan
=> Outlines how the project schedule will be developed, monitored and controlled.

5) Cost Management Plan
=> Describes how project cost will be estimated, budgeted and controlled.

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6) Quality Management Plan
⇒ Specifies the quality standards and how they will be achieved.

7) Resource Management Plan
⇒ Details how resources will be acquired, allocated and managed.

8) Communication Management Plan
⇒ Defines how information will be communicated among stakeholders.

9) Risk Management Plan
⇒ Identifies potential risks and outline strategies for mitigating them.

10) ~~Procedu~~ Procurement Management Plan
⇒ Describes how procurement will be managed, including contracts and vendor relationships.

11) Stakeholder Management Plan
⇒ Identifies all project stakeholders and outlines strategies for engaging and managing them.

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- Sample Template Sheet :-

Task ID	Task Name	Start date	End date	Duration	Assigned to	Status
1	Project initiation	01/06/2025	03/06/2025	3 days	John	Not started
2	Requirement Gathering	04/06/2025	10/06/2025	7 days	Sarah	Not Started
3	Design Phase	11/06/2025	20/06/2025	10 days	Mike	Not Started
4	Development	21/06/2025	30/06/2025	10 days	Alex	Not Started
5	Testing	01/07/2025	05/07/2025	7 days	Jane	Not Started
6	Deployment	06/07/2025	07/07/2025	2 days	Emily	Not Started.

1) Task ID : Unique identifier for each task.

2) Task Name : Brief description of tasks.

3) Start date : When the task is scheduled to begin

4) End date : When the task is scheduled to be completed.

5) Duration : Total time allocated for this task.

6) Assigned TO : Team members responsible for the task.

7) Status : Current progress of the task (e.g., Not started, In progress, completed).

Q.5 Describe earned value management technique in PM.

⇒

- Earned Value Analysis (EVA) is one of the key tools and techniques used in PM to have an understanding of how the project is progressing.
- EVA monitors the progress of the project based on its earnings or mony.
- Both, schedule and cost are calculated on the basis of EVA.
- Project control takes place against cost baseline using a technique called Earned Value.
- Key Components:

1) Planned Value (PV)

⇒ The budget for the work planned to be done by now.

- It shows how much work should have been completed by a certain point.
- Formula:

$$PV = BAC \times \text{Planned \% Complete}$$

2) Earned Value (EV)

⇒ EV is the actual progress of the task to the date of analysis.

- This is expressed as the percentage of the total efforts and/or resources expended

3) Actual Cost (AC)
⇒ The real cost spent on the work completed so far.
- It shows how much money the project has actually spent.
- Formula:

AC = Total actual costs upto now.

• Calculations:

1) Cost Variance (CV)
⇒ The amount that the project is above or below budget at the point of analysis.

$$CV = EV - AC$$

2) Cost Performance Index (CPI)
⇒ The amount that the project is above or below budget, relative to overall size of project.

$$CPI = \frac{EV}{AC}$$

3) Schedule variance (SV)
⇒ The amount that the project is ahead or behind schedule at point of analysis.

$$SV = EV - PV$$

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4) Schedule Performance Index (SPI)
=> The amount that the project is ahead or behind schedule relative to overall size of the project.

$$SPI = \frac{EV}{PV}$$

- Analysis :

a) A positive CV means the project is under budget. Negative means over budget. The CPI tells you how much above or below the budget it is, in percentage terms.
For e.g., CPI = 1.25 means the project is 25% below budget.

b) A positive SV means the project is ahead of schedule. Negative means behind schedule. The SPI tells you how much ahead or behind schedule the project is, in percentage terms.
For e.g., SPI = 0.9 means the project is ~~behind~~ 10% behind schedule.

Q.6 (Q.5a) May 24

⟹ Soln:-

AC = 250,000

PV = 241,000

EV = 252,000

1) Cost Variance = EV − AC

= 252,000 − 250,000

= Rs. 2,000

Positive CV means the project is under budget.

2) Schedule Variance = EV − PV

= 252,000 − 241,000

= Rs. 11,000

Positive SV means the project is ahead of schedule.

3) CPI = $\frac{EV}{AC}$

= $\frac{252,000}{250,000}$

= 1.008

CPI > 1 indicates cost efficiency.

4) $SPI = \frac{EV}{PV}$

$= \frac{252,000}{241,000}$

$= 1.0456$

SPI > 1 indicates schedule efficiency.

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Q.7 (Q.5 b) Dec 23

⇒ Solⁿ:-

AC = 45,000
PV = 35,000
EV = 31,000

∴ CV = EV − AC = 31,000 − 45000
= −14,000

→ Negative CV means over budget.

∴ SV = EV − PV = 31,000 − 35,000
= −4,000

→ Negative SV means the project is behind schedule.

∴ CPI = $\frac{31000}{45000}$ = 0.689

→ CPI < 1 indicates cost inefficiency.

∴ SPI = $\frac{31000}{35000}$ = 0.886

→ SPI < 1 indicates schedule delay.

Q.8 (May 23) (Q2.a)
⇒ Soln :-

AC = 35,000
PV = 27,000
EV = 31,000

∴ CV = EV − AC
= 31000 − 35000 = −4000
→ over budget

∴ SV = EV − PV
= 31,000 − 27,000
= 4,000
→ project is ahead of schedule

∴ CPI = $\frac{31000}{35000}$ = 0.886
→ cost inefficient

∴ SPI = $\frac{31000}{27000}$ = 1.148
→ schedule efficient.

Q.9 (May 22) (Q. 4 a)

→ Solⁿ :-

AC = 270,000

PV = 260,000

EV = 272,000

CV = 272000 − 270000 = 2000

SV = 272000 − 260000 = 12,000

CPI = $\frac{272000}{270000}$ = 1.007

SPI = $\frac{272000}{260000}$ = 1.046

CSI = CPI × SPI

= 1.007 × 1.046

= 1.053

→ Overall project health is Good.

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Q.10 What is scope creep? How does a formal change control system work in PM?

⇒

- Scope creep happens when extra work is added to a project.
- Scope creep is one of the biggest cause of project failure.
- It usually starts with a small change request, just a minor requirements of the project scope which is followed by one or more request and another.
- Causes of Scope Creep:

a) Not having a clear scope

⇒ Clarity is extremely important in any project.

- If you don't clearly define your scope at the beginning it can cause big problems down the line.

b) Not having client agreement

⇒ If the client isn't bought into the scope, they are likely to change their mind and the deliverables later on.

c) Poor estimation

⇒ Estimation is very difficult to get right. It's a challenge to be accurate at the beginning of the project when there are many unknowns.

d) Unrealistic schedule
⇒ Inaccurate time estimates lead to delays and added work.

e) Unsupervised changes
⇒ Changes made without proper approval disrupt time, cost and resources.

- Ways to Manage Scope Creep :

1) Be proactive
2) Prioritize
3) Be transparent
4) Analyze impacts
5) Embrace it

- Steps to avoid Scope Creep

1) Identify all stakeholders & understand their goals
⇒ You should do it even before you start working on the project.
- Identify visions, requirements and interests of all involved parties in advance it's a necessary step of project planning.

2) Clearly define the scope
⇒ It is another indispensable step of planning phase that needs close attention.
- It's important not only to define what is included in the project scope but also to list what is not included.

3) Plan room for changes in advance
⇒ Not all risks can be foreseen and prevented, and thus modifications are inevitable at times.
- That's why change control and management procedures should be clearly defined and documented; and room for possible amendments needs to be provided.

4) Take action as early as possible
⇒ Identify and address possible scope creep at its early stages.
- New suggestions from clients or sponsors, updated requirements or feedback from the team on time estimate change need to be heard, documented and communicated.

5) Know when to say no
⇒ It's tempting to implement all suggested features and functions, but it's rarely good for project delivery time; budget.
- Saying no to ~~unnecessa~~ unreasonably time consuming and expensive features and parts of work is crucial for delivering a quality product on time.
- Make sure you have a justification for saying no.

- How does formal change control system work in project?

1) Change is identified

⇒ Someone suggests a change to the project - this could be a new feature, a deadline shift or something else.

2) Change is tracked

⇒ The change is recorded and tracked through the entire project; from start to finish.

3) Change is reviewed

⇒ The project team or change control board looks at the change and decides if it's good for the project.

4) Decision is Made

⇒ The change is either

i) Approved

ii) Rejected

iii) Deferred

5) Change is documented

⇒ If the change is approved, it's officially added to the project plan, including updates to time, cost or scope.

6) Avoids disruption

⇒ This system helps keep things organized so the project doesn't go off track.

Q.11 - What is life cycle of project audit ?
What are responsibilities of project auditor?
What is essential for successful project audit?

⇒

- Project audit is a project evaluation technique designed to determine the true status of work performed on a project.
- The project audit life cycle :

1) Project Audit Initiation
⇒ Identify the main objective of the project.
- Write the project charter
- Get sign-off on the project charter.

2) Project Baseline definition
⇒ Review the original project plan.
- Establish reference points for performance comparision.

3) Establishing an Audit database
⇒ Gather project documents, reports, metrics, communication and stakeholder feedback.
- Collect quantitative and qualitative data.

4) Preliminary Analysis of the Project
⇒ Evaluate progress against the baseline.
- Identify variances, bottlenecks, risks and issues.
- Conduct ~~reviews~~ interviews with team members and stakeholders.

5) Audit Report Preparation
⇒ Summarize findings, strengths, weaknesses and non-compliance issues.
- Provide actionable recommendations.
- Grade or score project performance if required.

6) Project Audit Termination
⇒ Present the audit reports to stakeholders
- Achieve findings and close the audit.
- Use results for lessons learned and organizational improvement.

• Responsibilities of Project Auditor

1) Telling the truth and presenting unbiased, honest evaluation.
2) Maintaining objectivity, confidentiality and independence
3) Ensuring that both inclusions and omissions in the report are justified.
4) Selecting and managing an audit team.
5) Gathering and analyzing project data.
6) Conduting site visits and interacting respectfully with the project team.
7) Preparing a written report in a predefined format.
8) Sharing the report with the project team and ensuring follow-up on recommendations.

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• Essentials For a Successful Project Audit:

1) Clear audit objectives and scope.
2) Access to accurate and complete project documentation
3) Unbiased and skilled audit team.
4) Strong communication between auditors and stakeholders.
5) Top management support
6) Willingness of the project team to cooperate
7) Timely execution of audit activities
8) Actionable recommendations with follow-up plans.

Q.12 Briefly describe the purchasing cycle.

⇒

- The purchasing cycle, also known as procurement or procure-to-pay (P2P) cycle, is the structured process businesses follow to acquire good's and services.
- It ensures that purchases are made efficiently, cost-effectively and in compliance with organizational policies.
- Steps:

1) Needs Analysis

⇒ Identify what goods or services are required, such as raw materials, equipment or office supplies

2) Needs classification

⇒ Determine specific details like quantity, quality, brand preferences and delivery timelines.

3) Purchase Requisition

⇒ Create and submit a formal request for the needed items, often using standardized forms or procurement software.

4) Approval process

⇒ The requisition is reviewed and approved by the appropriate authority to ensure it aligns with budgetary and organizational goals.

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5) Supplier Selection
⇒ Evaluate potential suppliers based on criteria such as price, quality, reliability and delivery capabilities.

6) Purchase order Creation
⇒ Issue a purchase order to selected supplier, detailing the agreed-upon terms and conditions.

7) Order Fullfillment
⇒ The supplier delivers the goods or services as specified in the purchase order.

8) Receiving and Inspection
⇒ Upon delivery, the items are inspected to verify that they meet the required specifications and quality standards.

9) Record Keeping
⇒ Maintain detailed records of the entire purchasing process for auditing, compilances and future reference.

Q.13 Explain Project procurement management. What is the difference between contracting and outsourcing.

→

- The process of buying materials and obtaining services from vendors or dealers is called procurement.
- Material procurement is a very important part of project management.
- It involves the process of selecting vendors, establishing payment terms, stratergic vetting, negotiations contracts and actual purchasing of goods.
- Procurement is concerned with acquiring all of the goods, services and work that is vital to an organization.

• Process :-

- After project selection, project procurement management involves several important aspects such as procurement activities that are coordinated and managed throughout the life of a contract.
- For successful planning, execution and control of the procurement process, these components are crucial.
- Key components are :-

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1) Procurement Planning

⇒ The goods, services, or works that are required to be acquired are clearly defined.

2) Procurement Process

⇒ Identify potential suppliers or vendors who can be able to meet project requirements.

3) Contract Admission

⇒ Establish a legally enforceable document outlining the agreed-upon terms and conditions of an agreement between your project and its supplier.

4) Contract closure

⇒ Assuring that all deliverables and commitments laid down in this contracts are accomplished.

5) Risk Management

⇒ Determining possible threats in the procurement process, such as unreliable suppliers, changes to market prices or event ~~importan~~ improvements and reductions within project scope.

Procurement Management Plan:

- An organization's requirements, policies and general procedure for procurement are described in a procurement management plan for a specific project.
- Each organisation has a different level of complexity and detail when it comes to their procurement management plan.
- It should include:

1) **Procurement objectives & strategies**
⇒ Clearly defining the goals and approaches for acquiring goods and services needed for the project.

2) **Procurement processes**
⇒ Detailing the steps involved in procurement from identifying needs to contract closure.

3) **Roles and responsibilities**
⇒ Assigning tasks and responsibilities to team members involved in procurement, including the project manager, procurement officer and other stakeholder.

4) **Procurement methods**
⇒ Describing the specific methods or approaches to be used for procuring goods and services, such as competitive bidding, negotiation or direct purchase.

Contracting	Outsourcing
1) Hiring an external party to perform a specific task / service	1) Delegating an entire business function / process to a third-party.
2) Scope is Narrow, often short-term or project-specific.	2) Scope is Broad, involves long-term & comprehensive work
3) More control, client typically monitors and guides work.	3) Less control, third party manages operations independently.
4) Transactional agreement.	4) Strategic partnership
5) For e.g., Hiring a contractor to build a website or conduct training.	5) Outsourcing IT support or HR services to another company.
6) Separate from internal processes	6) Often integrated with internal processes

Q.14 What is contract? Explain types?
⇒
- Contracts are an essential part of procurement management.
- It creates a legal binding between the buyer and the seller.
- Contracts are necessary for project management as they provide relief on either side.
- Contract is an agreement between two parties in general.
- The agreement is made to procure goods and services required for the agreed project.
- This document needs to be prepared by the Project Manager.
- Guidelines to be considered while deciding on procurement contract:
  a) Complexity of procurement
  b) Location of procurement
  c) Availability of contractors
  d) Local governing regulations.

- Types
  - 1) Fixed-price contract
  - 2) Cost reimbursable
  - 3) Time & Materials contract

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1) Fixed - price Contracts :

- In this category, the contract involves a fixed price for a defined product or service or the result to be provided.
- Types :-

a) Firm fixed Price (FFP)

⇒ The price of the goods and services are set and never subjected to change unless the scope is changed and agreed mutually.

b) Fixed Price Incentive Fee

⇒ The price ceiling is set, and the seller needs to perform and full fill the contract requirements within that price.

- This type gives both the buyer and the seller some flexibility for performance with technical incentives.

c) Fixed price with Economic Price Adjustments

⇒ It is suitable when the contracts are executed in different countries and payment are made in a different currency.

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2) Cost reimbursable contracts

⇒ This type of contract involves cost reimbursement for the costs incurred during completion of the contractual jobs.

- Types :-

a) Cost Plus Fixed Fee

⇒ The seller gets all the allowable costs agreed in the contract.

- The seller also receives a fixed fee payment which is calculated as a percentage of initial estimated project costs.

b) Cost Plus incentive Fee

⇒ The seller gets the requirements for all the costs incurred on performing the work agreed in the contract.

c) Cost Plus Award Fee

⇒ In this type, the seller gets his/her legitimate reimbursements.

3) Time & Material Contracts

⇒ This a hybrid type of contract combining the feature of Fixed as well as Cost Reimbursable contracts.

- This is often used when contractual requirements is not known/prescribed.